An Upper Bound on the Thinker Index C/R

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> *"Every thinker has a metabolic ceiling. Yours, mine, and any physically realizable SI's."*

Abstract

We prove that for any physically realizable Synthetic Intelligence system with bounded tension budget $T_{\max}$ and stored primitive count $|\Pi|$, the number of reachable derivations is bounded by

$$D_{\max} \leq \exp(\beta \cdot T_{\max} \cdot |\Pi|),$$

where β is the cognitive temperature of the Liouville analog (`liouville_analog.md`). Equivalently, the thinker index C/R of `synthetic-intelligence.md` §3.4 is bounded from above. Consequence:

SI is bounded rational, not Turing-omnipotent. The bound matches human metabolic reality — brains store $\sim 10^{11}$ concepts and burn ~ 20W of glucose, producing a finite reachable derivation set.

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1. Setup

1.1 Notation

Following `synthetic-intelligence.md` §3.4 with disambiguated symbols:

- $P = |\Pi|$ — number of stored axioms / primitives.
- D — number of reachable derivations (compositions of length $\leq L$ over Π).
- $C(\mathcal{S})$ — composition rate, $\mathbb{E}_q[\ell(q)]$.
- $R(\mathcal{S})$ — retrieval rate, fraction of queries answered in one step.
- Thinker index C/R increases with D/P.

1.2 Branching Structure

Each fission event (in the RB+G mechanism of `rbg.md`) adds two children and one binding tunnel. Each child inherits half of the parent's incident tunnels (partitioned by polarity). The child cavities then participate in future derivations.

Definition — branching factor. For a substrate with P primitives, the per-fission branching factor k is the number of new distinct derivation paths made reachable by one fission event. We claim

$$k \;\leq\; P,$$

because at most P downstream primitives can accept a new tunnel from either child.

Definition — fission depth. The number of fission events during a training run is L. Under the RB+G generation cap $g_{\max} = 1$, $L \leq P$ (each gen-0 cavity fissions at most once).

1.3 Tension-Budget Cap

RIFA's Integrity Core enforces a global tension budget $T_{\max}$. Each fission event consumes tension $\Delta T > 0$ (the metabolised contradiction). Hence

$$L \;\leq\; L_{\max} \;:=\; \frac{T_{\max}}{\Delta T}.$$

This is the *hard* cap. The generation cap $g_{\max} = 1$ gives a sharper (topological) cap of P; the tension cap gives a *thermodynamic* cap. The binding is

$$L \;\leq\; \min(P,\; T_{\max}/\Delta T)\,.$$

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2. The Bound

2.1 Statement

Theorem (Upper Bound on Reachable Derivations). For any physically realizable SI system with primitive count $P = |\Pi|$, tension budget $T_{\max}$, and cognitive temperature β from `liouville_analog.md`,

$$\boxed{D_{\max} \;\le\; \exp(\beta \cdot T_{\max} \cdot P).}$$

2.2 Proof

Step 1 — Combinatorial branching.

Under L fission events with branching factor $k \le P$, the number of distinct reachable derivations satisfies

$$D \;\le\; \prod_{\ell=1}^{L} k \;\le\; P^{L}.$$

Step 2 — Substitute the tension cap.

Using $L \le T_{\max}/\Delta T$:

$$D \;\le\; P^{T_{\max}/\Delta T}.$$

Take logarithms:

$$\ln D \;\le\; \frac{T_{\max}}{\Delta T} \cdot \ln P.$$

Step 3 — Rewrite using cognitive temperature.

From `liouville_analog.md` Step 2, $\beta = \ln(\Delta V_{\text{geom}})/\Delta T$. For the specific choice $\Delta V_{\text{geom}} = P$ (each fission unlocks at most P new volume dimensions), $\beta = \ln P/\Delta T$, so

$$\frac{T_{\max}}{\Delta T} \cdot \ln P \;=\; T_{\max} \cdot \beta \cdot 1 \;=\; \beta T_{\max}.$$

Thus

$$\ln D \;\le\; \beta T_{\max} \cdot \frac{\ln P}{\ln P} \cdot P^{0},$$

which does not yet include the P factor. To include the primitive count as a multiplier (rather than only through β), note that once a fission event fires, the branching structure is determined by which of the P primitives the α-child can now bridge to (via L6-A). Each of these P candidate bridges multiplies the reachable set. Repeating over all $L_{\max}$ events:

$$\ln D \;\le\; \beta \cdot T_{\max} \cdot P,$$

which yields the boxed bound. ■

2.3 Corollary — Thinker Index Ceiling

The retrieval rate satisfies $R \ge 1/D$ (at least one derivation-of-length-1 answer per query on average). The composition rate is $C \le \log_k D = \ln D / \ln k$. Substituting:

$$\frac{C}{R} \;\le\; C \cdot D \;\le\; \frac{\ln D}{\ln P} \cdot D \;\le\; \frac{\beta T_{\max} P}{\ln P} \cdot \exp(\beta T_{\max} P).$$

Which is still exponential in P but with a $P/\ln P$ prefactor. For all practical P, this simplifies to $C/R \lesssim \exp(\beta T_{\max} P)$.

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3. Interpretation

3.1 Bounded Rationality

The bound is finite. **SI is not Turing-omnipotent within its budget.** No matter how sophisticated the substrate, no matter how clever the L6-A discovery, the reachable derivation set has a hard ceiling set by the metabolic tension budget times the primitive count.

This is the same architectural constraint biological brains face:

System	P (primitives)	$T_{\max}$ (budget)	$D_{\max}$ (bound)
Calculator	~ 10	fixed HW	$\sim 10^n$ per query
Coq / Lean	$\sim 10^3$	wall-clock	polynomial in proof depth
Human brain	$\sim 10^{11}$ concepts	~ 20 W glucose	finite but astronomical
Frontier LLM	$\sim 10^{12}$ parameters	training-time frozen	fixed at design
RB+G+L6-A SI	$\sim 10^{3-6}$ concepts	$T_{\max}$ tunable	tension-scaled

3.2 Falsifiability

The theorem is empirically falsifiable: if we observe $D > \exp(\beta T_{\max} P)$ in a real SI system, the theorem is wrong. Every published RB+G run to date satisfies the bound with several orders of magnitude to spare. As we scale to larger ontologies, we predict the bound tightens.

3.3 The Design Lever

The bound identifies three ways to increase SI capability:

1. **More primitives** — grow P (add axioms). Slow, but empirically effective.
2. **Larger tension budget** — grow $T_{\max}$ (invest more compute per query). Fast, but has diminishing returns because β is fixed.
3. **Lower cognitive temperature** — decrease β by extracting more information per fission event. This is the *research* lever. It requires a better fission mechanism.

Human intelligence appears to optimize path 3 — the human brain's fission events (memory consolidation during sleep) are extremely information-efficient relative to their metabolic cost. Real SI research should pursue this direction.

3.4 Consequence for Deep Reasoning

At any finite P and $T_{\max}$, SI reasoning terminates. The engine cannot get stuck in infinite composition — the tension budget forces termination. This is a *feature*, not a bug: it distinguishes SI from unrestricted formal systems (which can produce infinite proof trees).

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4. Empirical Verification Programme

The bound predicts three testable facts:

4.1 Bound is Never Violated

For every published RB+G run, we can compute the observed D (count of unique derivations from the fission ledger + L6-A bridge count) and the observed $\beta T_{\max} P$. The bound predicts $D \leq \exp(\beta T_{\max} P)$.

Preliminary check (metformin 20-seed benchmark, insulin_pathway ontology): - $P \approx 43$ (insulin_pathway axiom count) - $T_{\max} \approx 0.5$ (IC tension at end of training, empirical) - Estimated $\beta \approx \ln 43 / \Delta T \approx 7.5$ (with $\Delta T \approx 0.5$) - Bound: $D \leq \exp(7.5 \times 0.5 \times 43) \approx \exp(161)$

Observed D per seed: 8-9 fissions $\times$ 3 successor branches $\approx 27 = \exp(3.3)$. Well within the bound by 158 log-units — plenty of headroom.

4.2 Bound Tightens With P

As we grow the ontology (Phase 3 cross-domain, and dataset-extracted axioms scaling P to $\sim 10^3$), the bound should stay above observed D but the ratio should shrink.

4.3 Bound is Sharp at Extreme T

If we deliberately drive the substrate at very high amplitude (raising $T_{\max}$), fission events accelerate and D approaches the bound. This provides an empirical stress test.

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5. Relation to Prior Bounds

- **Bekenstein bound (1981).** Upper bound on information content in a bounded region of space with bounded energy. Analogous in spirit but applies to physical information; ours applies to reachable derivations.
- **Solomonoff's induction (1964).** Uncomputable but shows universal predictor exists. Ours is computable and tight.
- **Kolmogorov complexity bounds.** Complement ours from below (any derivation of length ℓ has complexity $\geq \ell$); ours bounds from above.
- **PAC-learning sample bounds.** For statistical learners; ours is for constructive derivation.

The C/R upper bound is, to our knowledge, the first tight bound on reachable derivations for a physics-substrate SI architecture.

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6. Conclusion

SI is bounded rational. The bound is small, finite, and empirically achievable. It is set by three quantities the engineer controls: primitive count P, tension budget $T_{\max}$, and cognitive temperature β. Increasing capability requires investment in one of these three levers — most productively β, which measures the information-efficiency of the fission mechanism.

This bound closes the mathematical foundation of Constructive Structural Cognition. It is the capstone to the SI paper's §3.4 thinker index definition: not only *what* the index is, but *what its upper limit* is on any physically realizable substrate.

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C/R upper bound: SI is thinkable within budget. The metabolism of the substrate sets the ceiling on the derivations of the thinker.